Testing

If it isn’t tested, it isn’t done

Testing

- What is Testing?
- Testing Methodologies
- Website testing
- Testing checklists

What is testing?

What is the purpose and what are the qualities of testing?

Testing

- Investigation into product quality
- Assessment of investigation findings
- Objective and independent, as far as possible
- Provides stakeholders with information about
  - Product quality
  - Business risks of product deployment
- Validates and verifies that a product
  - Meets the user needs
  - Works robustly as expected
  - Can be deployed

Testing Methodologies

You have to have a system

Testing Methodologies

- Static and Dynamic testing
- Box models
- Automated testing

Static and Dynamic

Truth and Worth


Static
Dynamic


Static
Dynamic

Walkthrough

Static

- Guided demo of the product and discussion with users
- Gain feedback
- Familiarises the stakeholders with the product

Inspection

Static

- Inspectors systematically look for defects
- Planning: inspection is planned by the moderator
- Overview meeting: developer describes product background
- Preparation: Inspectors identify possible defects
- Inspection meeting:
  - Developer demos the product in a structured manner
  - Inspectors point out the defects for every part
- Rework: developer makes changes according to an action plan
- Follow-up: changes are checked to ensure compliance

Code Review

Static

- Systematic examination of the source code
- Find and fix mistakes overlooked during development
- Improving overall quality and developer skills


Static
Dynamic


Static
Dynamic

Dynamic Testing

Dynamic

- Executing programmed code with a given set of test cases
- May begin before the product is complete
  - To test particular sections of code
  - Can be applied to discrete functions or modules

Test Case

Dynamic

Project Name:

Test Case Template

Test Case ID: Fun_10

Test Priority (Low/Medium/High): Med

Module Name: Google login screen

Test Title: Verify login with valid username and password

Description: Test the Google login page

Test Designed by: <Name>

Test Designed date: <Date>

Test Executed by: <Name>

Test Execution date: <Date>

Pre-conditions: User has valid username and password
Dependencies:

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Navigate to login page	User= example@gmail.com	User should be able to login	User is navigated to	Pass	
2	Provide valid username	Password: 1234		dashboard with successful		
3	Provide valid password			login		
4	Click on Login button					

Post-conditions:
User is validated with database and successfully login to account. The account session details are logged in database.

http://www.softwaretestinghelp.com/test-case-template-examples/


Static
Dynamic


Static
Verification
Dynamic
Validation

Truth

Worth

Product vs. Specification

Fitness to operational purpose

Truth

Worth

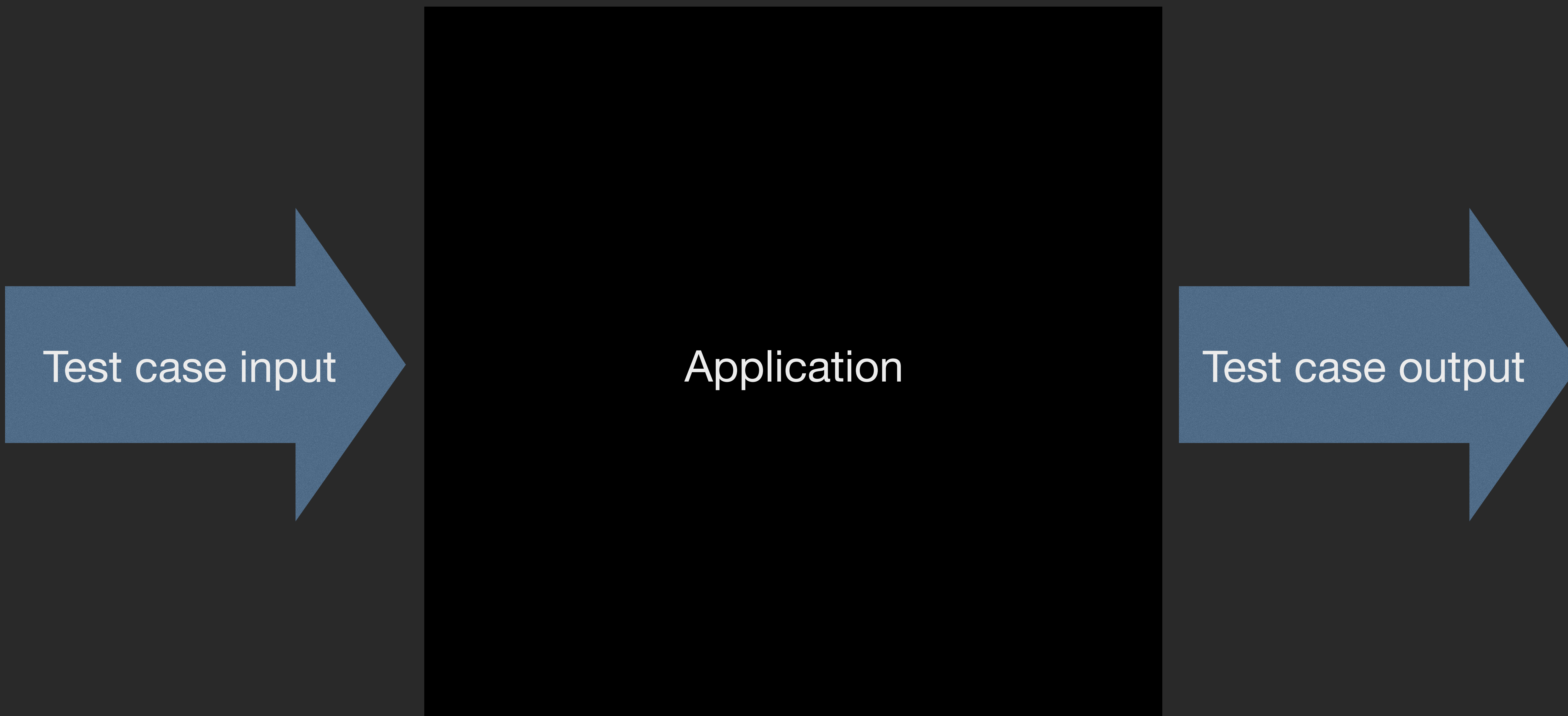
Product vs. Specification

Fitness to operational purpose

Box Models

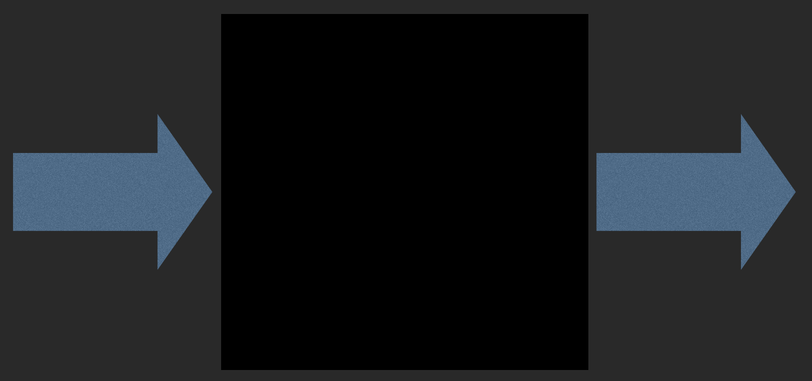
Now you see it, now you don’t

Black Box

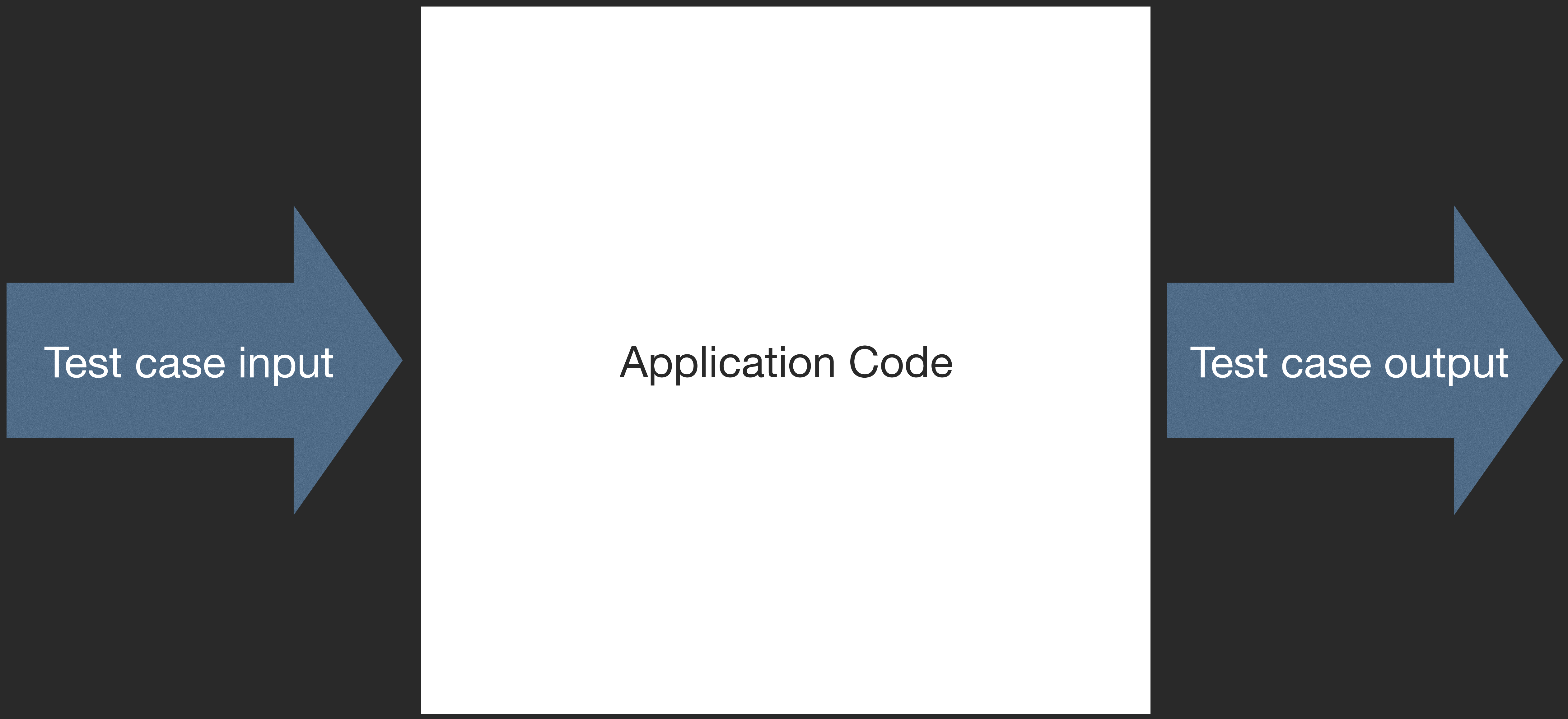


Black Box

- Test functionality without knowledge of internal implementation
- Tester is only aware of what the software is supposed to do
- No knowledge of how the software works internally is used
- Tests are typically based on Test Cases

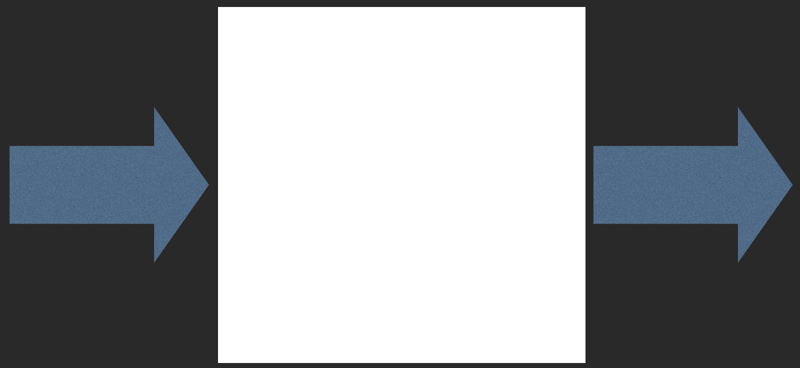


White Box



White Box

- Tests internal structures or workings of a program
- Tests inputs
- Tests expected outputs
- Design of test cases makes use of
  - Internal perspective of the system
  - Programming skills
- Might not detect unimplemented features
  - i.e. missing requirements



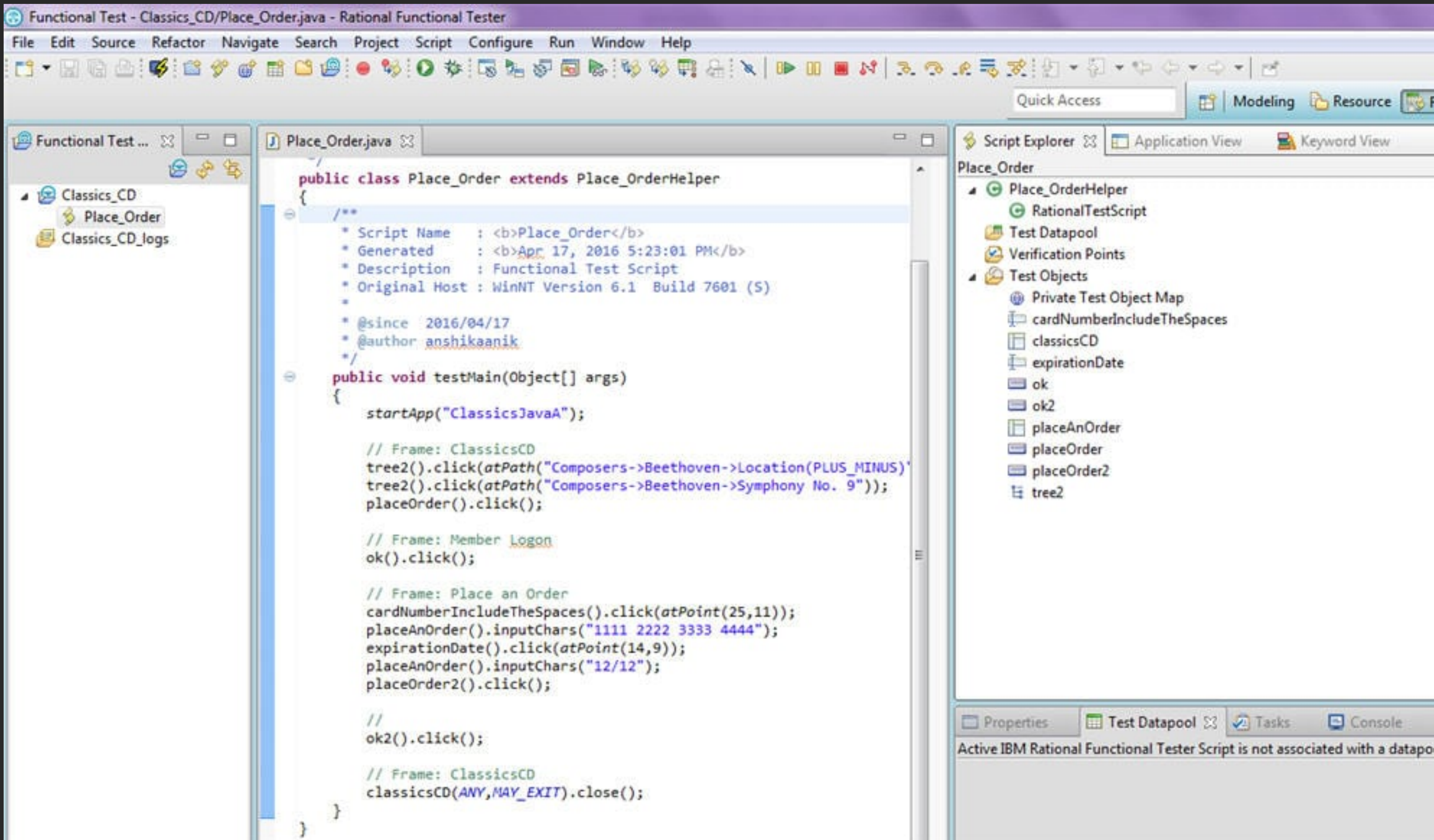
Automated testing

A robot did it

Automated Testing

- The use of automation to apply test cases
- Range from basic Linear to more sophisticated BDD approaches
- Reduces the number of test cases to be run manually
- Reduces overall time and associated cost
- Eases testing of multi-lingual products
- Provides consistent reproducibility, which is critical for Agile
- However, complements rather than replaces human testing

Automated Testing Tools



IBM Rational Function Tester

Automated Testing Tools

```java
import org.openqa.selenium.By;
import org.openqa.selenium.Keys;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.support.ui.WebDriverWait;
import static org.openqa.selenium.support.ui.ExpectedConditions.presenceOfElementLocated;
import java.time.Duration;

public class HelloSelenium {

    public static void main(String[] args) {
        WebDriver driver = new FirefoxDriver();
        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
        try {
            driver.get("https://google.com/ncr");
            driver.findElement(By.name("q")).sendKeys("cheese" + Keys.ENTER);
            WebElement firstResult = wait.until(presenceOfElementLocated(By.cssSelector("h3>div")));
            System.out.println(firstResult.getAttribute("textContent"));
        } finally {
            driver.quit();
        }
    }
}
```

Selenium

Testing in Agile

I like your manifesto, I’ll put it to the test-o

Agile Testing Manifesto

Growing Agile: A Coach's Guide to Agile Testing (2015), Samantha Laing and Karen Greaves

Website testing

Key considerations for web developers

Testing all aspects

- Functionality testing
- Security testing
- Integration testing
- Performance testing
- Compatibility testing
- Usability testing

Functionality Testing

- Make sure all functions are working robustly
- Covers a range of feature types
  - Forms
  - Cookies
  - Database integrity
- Must ensure that test cases cover boundary conditions

Security Testing

- Critically important for websites
- Test bot activity detection
- Check for unwanted direct access to server file structure
- Prevent SQL script injection

Integration testing

- Test that all system components work together properly
  - Architectural level, e.g. client/server, APIs and frameworks
  - Unit level, that high-level functional parts work together
  - Lower-levels, good integration at the function/module level
- Particularly important in a team development context

Performance Testing

- Helps to determine product performance across use cases
  - Stress testing
  - Scalability testing
  - Load testing
- Test on a range of deployment platforms

Compatibility Testing

- Compatibility of your web application is crucial
- Standards compliance
- Accessibility and the legal compliance
- Browser compatibility
- Operating system compatibility
- Context compatibility
    - e.g. mobile settings

Usability Testing

- Website should comply with usability standards
- Test against global conventions and Web standards
- Keep in mind important aspects:
  - Cognitive load
  - Support of user goals
  - Information design
  - Navigational design
  - Visual design
- Critically, this includes all users
  - i.e. Accessibility testing

Testing checklists

Make a list and check it twice

Testing checklist

- Use of checklist has been shown to improve outcomes
- Example testing checklists
  - Editor
  - Designer
  - Developer
  - SEO
  - Network Administration
  - Security
  - Legal

Editor checklist

- Spelling, Grammar and Punctuation
- Forms
  - Can the flow be improved?
  - Do you get stuck?
  - Are the instructions accurate?
  - Does the completed form get sent to the right people or person?
- Content
  - Why would I visit this page?
  - Is the content ready for visitor?
  - Does the page address the audience?
- Speed
  - Check the size of your page sizes and their load time

Designer checklist

- Compatibility
  - Do the pages render well in common browsers?
- Images
  - Do all images define and useful *alt* attribute?
  - Do images display correctly?
  - Have images been optimised for website use
- Font availability and support
  - Are the fonts available to the intended users?
  - Do font substitutes preserve the website’s presentation?

Developer checklist

- Check code against standards before release
  - Does the west meet W3C standards?
- Optimise page downloads
  - Have scripted been minified?
- Unavailable content
  - Have helpful custom 404 pages been defined?
  - Can visitor find something else of use on the website instead?
- Check deployed website
  - Do all links work after deployment?
  - Do all processes work after deployment?

SEO checklist

- Meta data
  - Do all pages have a unique title tag?
  - Do all pages have a meaningful meta description?
- Make sure your analytics tools are ready to collect data
  - Have analytics scripts been included?
  - Have analytics metrics been specified?
- Social Media Integration
  - Are all social media links working correctly?
  - Have the correct social media plugins been integrated?
- Ensure any new version of the website replaces the old version
  - Do all new links, media, etc. remap correctly?

Network administration checklist

- Monitoring
  - Are processes in place to monitor site availability?
  - Are important pages monitored at a detailed level?
- Traffic Loads
  - What might happen under a sudden heavy traffic load?
  - Does your service provider have DDoS mitigation provision?
- Backup System
  - What happens if the server goes down?
  - Is the backup system is configured and verified properly?
  - Have the website failover and failback processes been tested?

Security checklist

- Cyberattack
  - Is the website scripted to resits attack?
    - For example, Script Injection and Cross Site Scripting
  - Is received data verified and cleansed on client and server sides?
- Protected Pages
  - Are protected areas protected?
  - Are the approached used robust?
- Secure Certificate (if required)
  - All all certificates in place and checked?

Legal checklist

- Accessibility
  - Are the Equality Act's, Provision of a Service, requirements met?
  - Does the website meet W3C Web Context Accessibility Guidelines?
  - Has the website been tested with disabled users?
  - Have checks been made for accessibility in the case of:
    - Eyesight conditions: Colourblindness, cataracts, tunnel vision?
    - Motor conditions: Arthritis, Parkinson's Disease, Limb damage or loss?
    - Cognitive conditions: Autism, Acquired Brain Injury, Dementia, Dyslexia?
- Data protection
  - Are users notified of cookies and other forms of data storage?
  - Is personal data stored securely?
  - Is personal data stored confidentially?
- Have legal requirements been met for all legal jurisdictions required?

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